

Brazilian E-Commerce Public Dataset

5. Exploratory Data Analysis Report

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1. EDA Scope

- EDA was conducted on order_level_clean.csv (10,000 rows) and item_level_clean.csv (11,383 rows).
- Six domains analysed: order status and volume, geographic distribution, delivery performance, product and category analysis, payment behaviour, customer satisfaction and retention.

2. Order Status and Volume

- 97.1% of orders carry delivered status -- dataset is skewed towards completed transactions.
- Remaining statuses: shipped 1.0%, canceled 0.7%, unavailable 0.6%, others under 0.3%.
- Monthly order volume shows a clear upward trend through 2017-2018, with a spike in November 2017 (Black Friday) and a sharp tail-off as the dataset window closes.
- The volume growth pattern confirms the platform was in active expansion during the observed period.

3. Geographic Distribution

Rank	City	Share of Orders	Note
1	Sao Paulo (city)	15.8%	Largest single-city contributor
2	Rio de Janeiro	6.7%	Second largest
3	Belo Horizonte	2.9%	Third largest
4	Brasilia	2.2%	National capital
5	Campinas	1.5%	Major SP state satellite city

- Top 5 cities represent ~29% of all orders -- demand is heavily urban-concentrated.
- At state level, SP leads by a large margin, followed by RJ and MG -- mirrors Brazil's GDP distribution.
- Any logistics disruption in Sao Paulo state would have an outsized effect on platform-wide performance.

4. Delivery Performance

Metric	Value
Mean delivery time	12.0 days
Median delivery time	10.0 days
Fastest delivery	0 days (same-day)
Slowest delivery	186 days
On-time delivery rate	92.3%
Late delivery rate	7.7%

- The 2-day gap between mean (12d) and median (10d) indicates a right-skewed distribution inflated by a small number of extreme delays.
- 92.3% on-time rate reflects the platform consistently padding delivery estimates to protect satisfaction scores.
- The 7.7% late deliveries are disproportionately associated with 1-star reviews based on the bimodal distribution pattern.

5. Product and Category Analysis

Rank	Category	Revenue (BRL)	Observation
1	Health and Beauty	141,728	Lifestyle leader
2	Watches and Gifts	136,689	Gifting category
3	Bed / Bath / Table	122,963	Home essentials
4	Sports and Leisure	110,422	Active lifestyle
5	Computers and Accessories	106,869	Only tech category in top 5
6	Furniture and Decor	105,844	Home decor

7	Garden Tools	103,112	High unit volume
8	Auto	97,055	Automotive accessories
9	Cool Stuff / Novelties	88,413	Gifting and impulse
10	Construction Tools	82,222	DIY segment

- Top 3 categories account for ~25% of total item revenue (R\$1,581,151).
- Computers and Accessories is the only tech category in the top 5 -- this is primarily a lifestyle and home goods marketplace.
- Average basket size is 1.15 items per order -- almost all purchases are single-item transactions with no visible bundling behaviour.

6. Seller Analysis

- 1,654 unique sellers listing an average of ~4 products each across 6,747 distinct products.
- Top 20% of sellers (~331) drive 72.2% of total item revenue -- the bottom 80% contribute less than 28%.
- Top 10 sellers each generated R\$16,000 to R\$25,000, clustered tightly with no single dominant outlier.
- Two distinct seller strategies observed: high-volume low-price (201 items at avg R\$106) vs. low-volume premium (57 items at avg R\$406).
- Sao Paulo state dominates seller geography, consistent with hosting Brazil's largest commercial infrastructure.

7. Payment Behaviour

Payment Method	Share	Observation
Credit card	76.5%	Dominant -- spans all income segments via installments
Boleto	19.7%	Cash slip -- reflects unbanked/underbanked population
Voucher	2.3%	Promotional credit -- negligible volume
Debit card	1.5%	Very low adoption on this platform

- 48.9% of orders settled in a single installment; average when split = 2.9 installments -- short-term convenience, not long financing.
- Effectively a two-method marketplace -- credit card and boleto together cover 96% of all transactions.

8. Customer Satisfaction

Score	Share	Observation
5 stars	58.0%	Majority positive
4 stars	18.9%	Satisfied
3 stars	6.5%	Neutral
2 stars	5.1%	Dissatisfied
1 star	11.4%	Strong negative

- Average review score: 4.09 / 5.00 across 9,912 reviewed orders.
- Bimodal distribution -- customers either rate 5 stars or jump to 1 star. Middling scores are rare, pointing to polarised outcomes.
- 76.9% of reviewers gave 4 or 5 stars -- positive sentiment is the clear majority.

9. Customer Retention

- 9,956 unique customers placed 10,000 orders -- only 43 customers (0.4%) placed more than one order.
- Average orders per customer is 1.00. This is an acquisition-only platform with no organic repeat purchase behaviour.
- High spend does not predict satisfaction: 4 of the top 10 highest-spending customers gave 1-star reviews.
- The highest single-order spend was R\$7,274 -- almost certainly a single high-value item such as furniture or a large appliance.